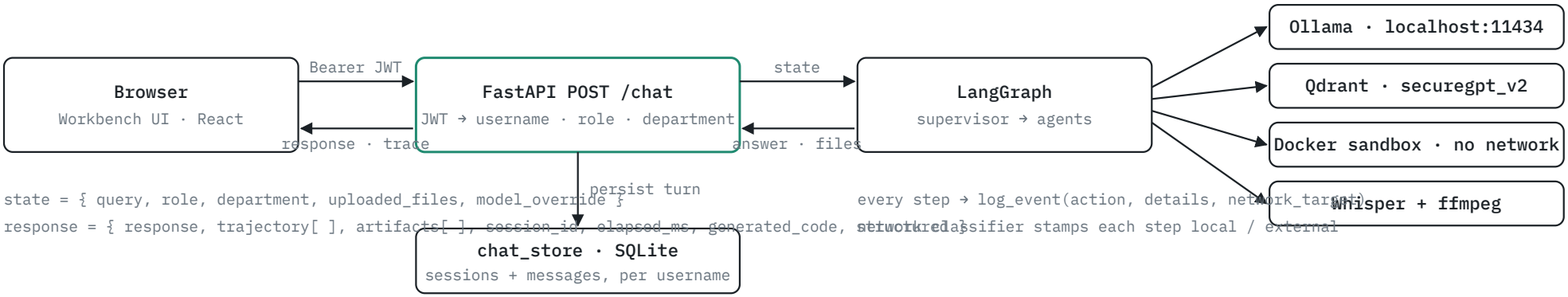
What happens between a question and its answer

Every hop a query takes, drawn from the code that runs today: the request boundary, the supervisor's decision ladder, the three-stage retrieval cascade, the inside of one hybrid search, and the three specialist agents. Dashed boxes are LLM calls; teal marks the security boundary and the default path; amber numbers are the gates.

01 Request boundary

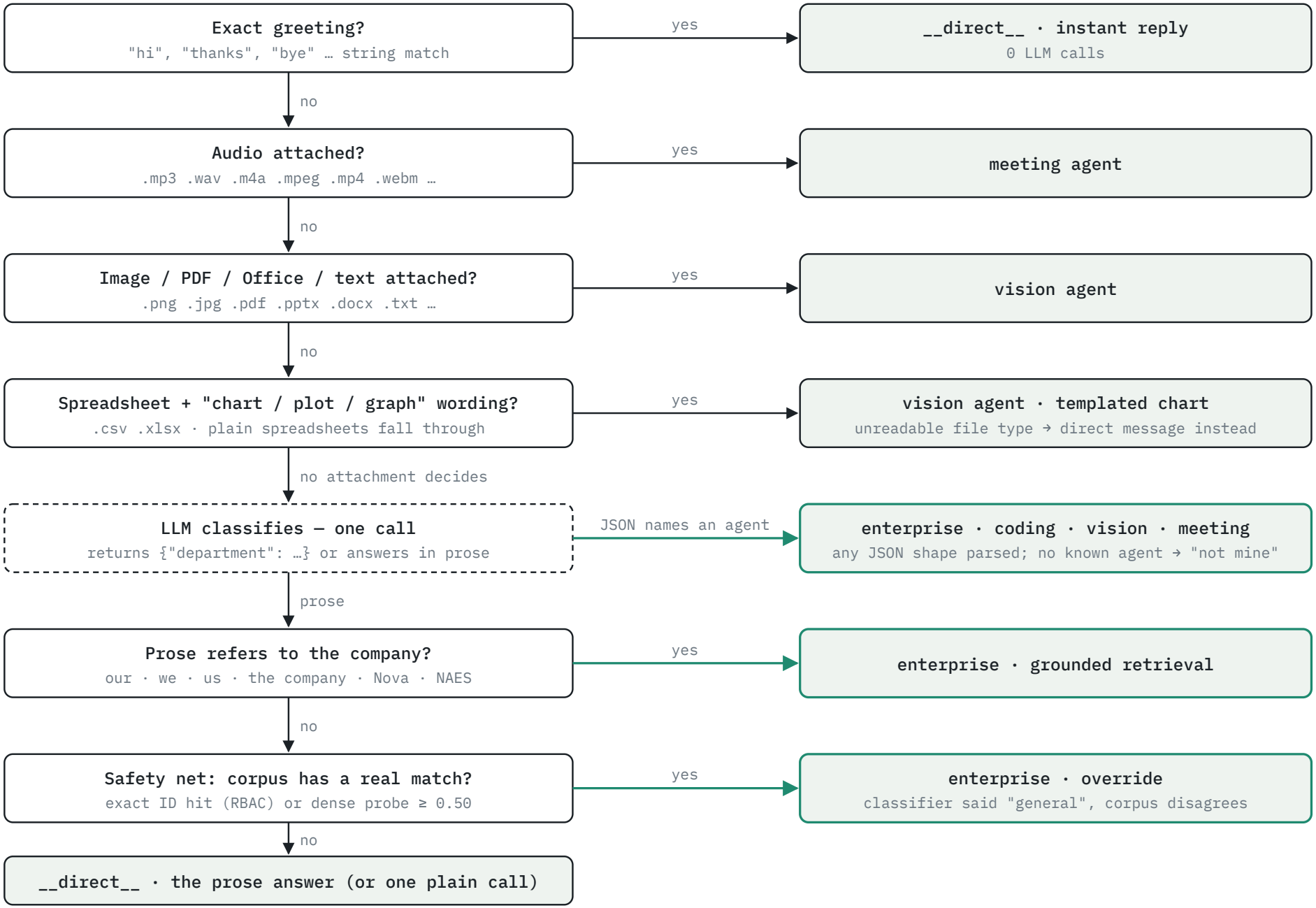
The UI never talks to a model or a database. One authenticated POST enters FastAPI, which turns the JWT into an identity, runs the graph, stores the turn, and returns the answer with its full trajectory.



Identity is decided once, at the API boundary, and travels inside the graph state; every retrieval below filters on it. The trajectory the UI shows under "Trace" is the same event list written to `logs/session.jsonl`.

02 Supervisor — who handles this?

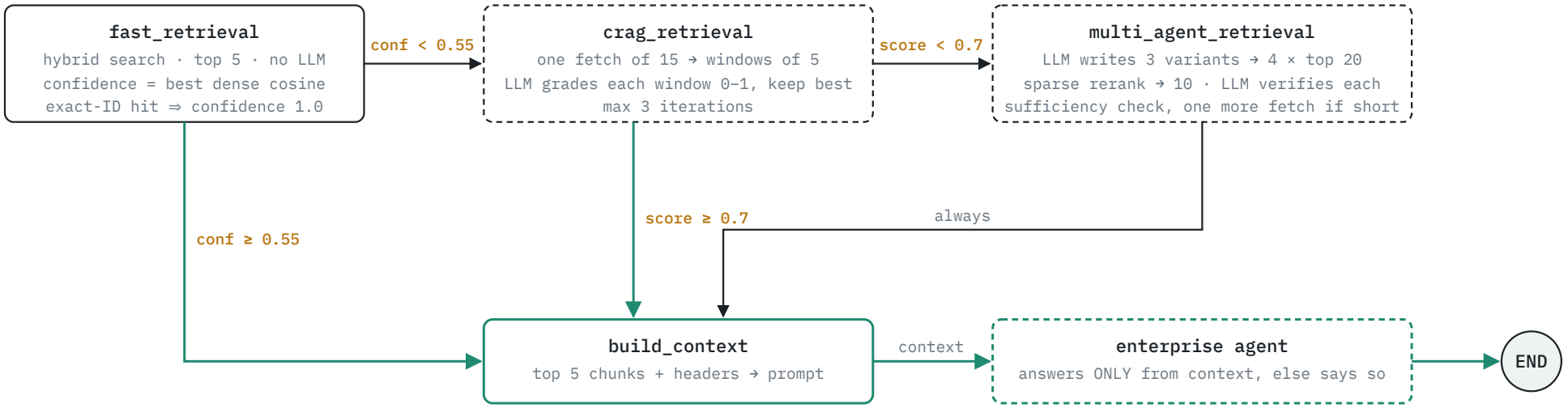
Cheap, deterministic checks run before any model is consulted. Only when they don't decide does a single LLM call classify the query — and even its "answer directly" verdict is second-guessed against the corpus.


Exact greeting?
"hi", "thanks", "bye" … string match
yes
__direct__ · instant reply
0 LLM calls
no
Audio attached?
.mp3 .wav .m4a .mpeg .mp4 .webm …
yes
meeting agent
no
Image / PDF / Office / text attached?
.png .jpg .pdf .pptx .docx .txt …
yes
vision agent
no
Spreadsheet + "chart / plot / graph" wording?
.csv .xlsx · plain spreadsheets fall through
yes
vision agent · templated chart
unreadable file type → direct message instead
no attachment decides
LLM classifies — one call
returns {"department": …} or answers in prose
JSON names an agent
enterprise · coding · vision · meeting
any JSON shape parsed; no known agent → "not mine"
prose
Prose refers to the company?
our · we · us · the company · Nova · NAES
yes
enterprise · grounded retrieval
no
Safety net: corpus has a real match?
exact ID hit (RBAC) or dense probe ≥ 0.50
yes
enterprise · override
classifier said "general", corpus disagrees
no
__direct__ · the prose answer (or one plain call)

Attachments decide before any model runs because a file's type is a fact, not a judgement. The company-reference rule and the corpus probe exist because a 3B classifier calls "what did our risk report flag?" general knowledge about one time in five; both catch it deterministically.

03 Enterprise cascade — retrieve as cheaply as the query allows

Every enterprise query starts on the fast path. It escalates only when the evidence is weak, and each stage is strictly more expensive than the last: no LLM, then one grading call per window, then rewrites plus per-chunk verification.

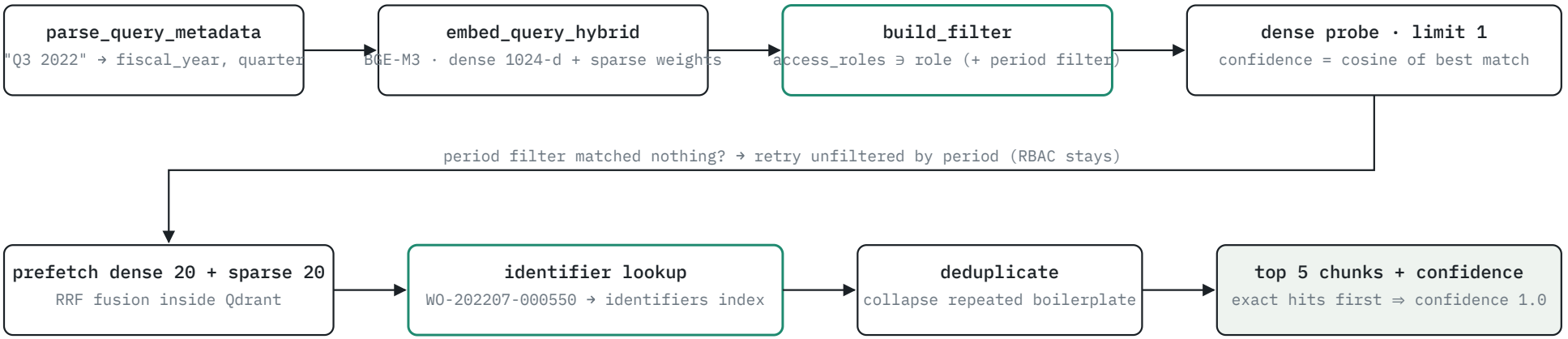


Coding, vision and meeting bypass this cascade entirely: the supervisor sends them straight to their agent (figure 05).

The gate on the fast path is the raw dense cosine of the single best match, never the fused RRF score — RRF is rank-based and looks confident even when nothing relevant exists. The same 0.50 floor that gates the supervisor's safety net is what makes 0.55 a meaningful step above it.

04 Inside one `retrieve()` call

Every stage above calls the same function. It embeds once, filters by identity inside Qdrant, fuses dense and lexical rankings, and puts exact identifier matches ahead of everything.

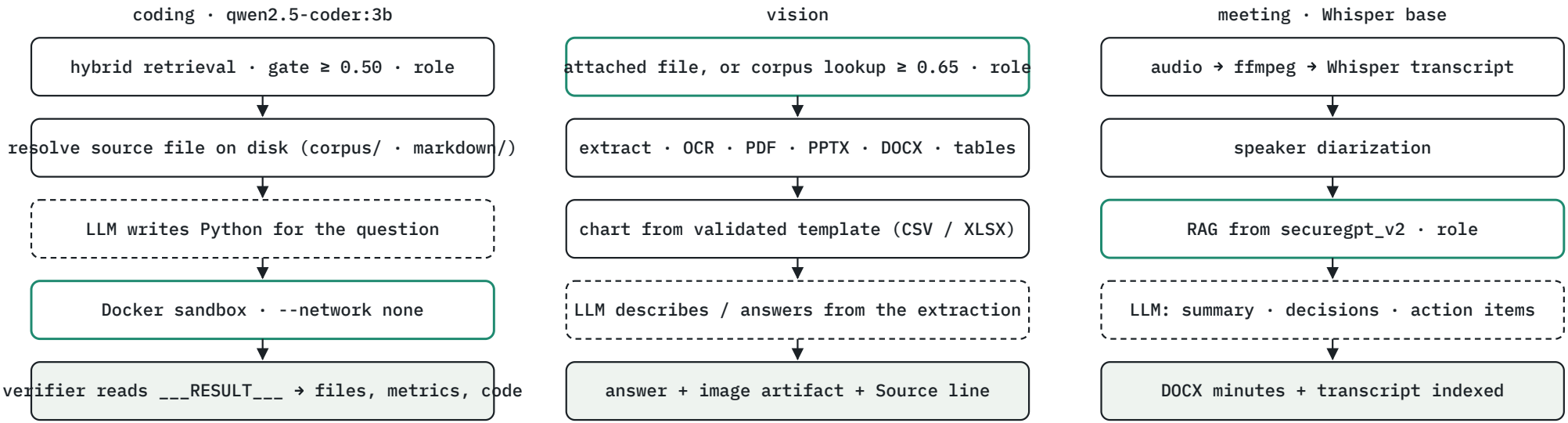


The RBAC filter is passed to every Qdrant call here — probe, both prefetches and the identifier lookup — so no stage can see a chunk the caller's role cannot.

Dense similarity finds meaning; the sparse (lexical) vector finds wording; the identifier index finds record codes that neither reliably surfaces — the 10-row eval showed a work-order ID retrieved by a broad question and missed by the exact one, which is why the lookup exists.

05 Specialist agents

The three agents that don't use the cascade. Each still retrieves under the caller's role when it needs the corpus, and each ends in the same response shape — answer, trajectory, artifacts.



All three return through the same FastAPI response: generated files are published as artifacts, the code as generated_code, verified numbers as structured.

Coding never sees the network — the sandbox runs with `--network none` and only the verified `___RESULT___` block is trusted. Vision retrieves a corpus document only when the wording asks for one; a "chart this" request with an attachment never touches Qdrant.

deterministic step · LLM call (Ollama, local) · security boundary or default path · **0.55** gate, tunable by env var

06 The gates, as configured today

Every number in the figures is read from configuration at start-up; none is fixed in a node. Values below are the current defaults.

SETTING	VALUE	WHAT IT DECIDES
DENSE_RELEVANCE_THRESHOLD	0.50	Floor for "the corpus has a real match" — supervisor safety net and the coding agent's gate. Unrelated queries measure ~0.45–0.46; genuine ones 0.59+.
FAST_PATH_CONFIDENCE_THRESHOLD	0.55	Fast path accepted without grading; below it, CRAG runs.
CRAG_SUFFICIENT_SCORE	0.7	Best-window grade needed to stop escalating to multi-agent retrieval.
CRAG_MAX_ITERATIONS · CRAG_WINDOW	3 · 5	Windows graded per query and chunks per window; one fetch of 15 covers all three.
TOP_K · MAX_CONTEXT_CHUNKS	5 · 5	Chunks returned per retrieve() and forwarded into the prompt.
VISION_SOURCE_THRESHOLD	0.65	Higher bar for the vision agent to open a whole corpus document on its own.
IDENTIFIER_LOOKUP_LIMIT · IDENTIFIER_MIN_LEN	3 · 5	Exact-ID chunks reserved at the front of the context; shortest token treated as a code.
RBAC_SCOPE	role	Filter on the chunk's access_roles only. "department" would also admit any chunk from the caller's department.
QDRANT_COLLECTION	securegpt_v2	763,497 chunks · named vectors dense + sparse · payload indexes on access_roles, department, identifiers, text.

Source of each figure: api/routes/chat.py (01) · graph/agents/supervisor.py (02) · graph/workflow.py, graph/nodes.py (03) · retrieval/retrieve.py, retrieval/identifier_lookup.py, retrieval/filter_builder.py (04) · graph/agents/coding_agent/agent_graph.py, graph/agents/vision.py, meeting_agent/workflow.py (05) · retrieval/config.py (06).